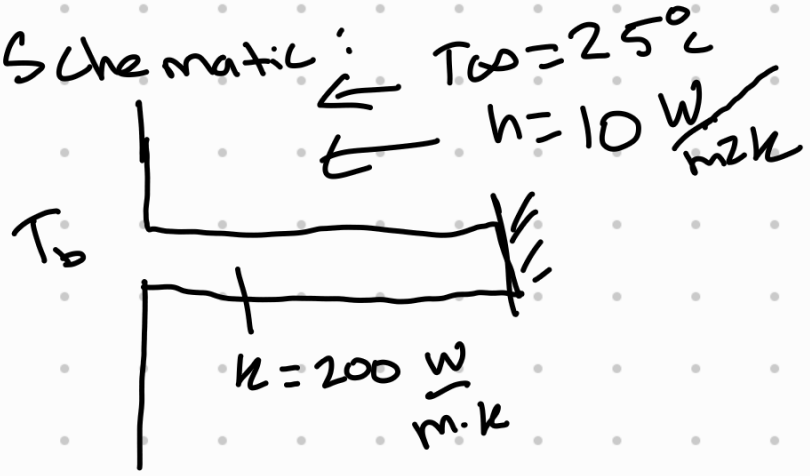
MAE 3240 HW 5 due 3-10-2026

#1 Known: $r = 1mm$ $L = 5cm$ $T_\infty = 25°C$
$h = 10 \frac{W}{m^2 K}$ $k = 200 \frac{W}{m \cdot K}$, tip thermally insulated.

Assumptions: 1D heat flow, uniform crosssection, tip area small compared to surface area

Schematic:



a) $T_b = 125°C$ → find fin temp @ $X = L/2$ and $x = L$

case B in table 3.4

$\theta = T - T_\infty$

$\theta_b = T_b - T_\infty$

$$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh(mL)}$$

$$m = \sqrt{\frac{hP}{kA_c}}$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{10 \frac{W}{m^2 \cdot K} * 2 * 0.001m \pi}{200 \frac{W}{m \cdot K} \cdot (0.001m)^2 \pi}} = 10 = m$$

$\theta_b = 125°C - 25°C = 100°C$

$$\frac{\theta(x=0.05m)}{100°C} = \frac{\cosh[10(0.05m - 0.05m)]}{\cosh(10 * 0.05m)}$$

$\theta = T - T_\infty = T - 25°C = 88.68$

$$\boxed{T(x=L) = 113.68°C}$$

$$\frac{\theta(x=0.025m)}{100°C} = \frac{\cosh(10(0.025 - 0.05))}{\cosh(10 * 0.05)}$$

$$\boxed{T(x=\tfrac{L}{2}) = 116.46°C}$$

b) Find fin effectiveness ε_f and fin efficiency η_f according to table 3.4 in textbook.

$$\varepsilon_f = \frac{kP}{hA_c}\tanh(mL), \quad \eta_f = \frac{q_f}{hA_f\theta_b}$$

$$\varepsilon_f = \frac{q_f}{hA_c\theta_b} \leftarrow q_f = M\tanh mL \leftarrow \text{from textbook}$$

$$q_f = \sqrt{\overset{h}{10\frac{W}{m^2K}} * \overset{P}{0.002\pi m^2} * \overset{k}{200\frac{W}{mK}} * \overset{A_c}{(0.001m)^2\pi}} * \overset{\theta_b}{100°C}\tanh(mL)$$

$q_f = 0.628\tanh(10 * 0.05m)$

$q_f = 0.29W$ ← not fully sure on units

$$\varepsilon_f = \frac{0.29}{10\,\frac{W}{m^2 \cdot K} * (0.001m)^2 \pi * 100}$$

$$\boxed{\varepsilon_f = 92.3}$$

$$\eta_f = \frac{0.29}{\underset{h}{10\,\frac{W}{m^2 \cdot K}} * \underset{\theta_b}{100} * \underbrace{0.002 * \pi}_{P} * \underset{L}{0.05m}}$$

$A_f = PL$
$= 2\pi RL$

$$\boxed{\eta_f = 0.95}$$

c Find η_f based on table 3.5. Is it a good approximation compared to part b?

$$\eta_f = \frac{\tanh(mL_D)}{mL_D}$$ ← from table 3.5 in textbook

$$L_D = L + \frac{R}{2} = 0.05m + \frac{0.001}{2} m$$

$$\eta_f = \frac{\tanh(10 * 0.0505)}{(10 * 0.0505m)} = \boxed{0.922 = \eta_f}$$

this result is pretty close to the actual from part b, so this is a good approximation.

d) Find thermal resistance of the fin R_f

$$R_f = \frac{\theta_b}{q_f} \qquad \frac{100\,K}{0.29} = \boxed{344\,\frac{K}{W} = R_f}$$

e) Fins form 5x5 array mounted on base plate s = 3cm.

Find overall efficiency η_o, and thermal resistance of the fin array R_o

Find how much heat is dissipated through the unified surface vs fins.

$$1 - \frac{N A_f}{A_{total}}(1-\eta_f) = \eta_o$$

$$A_{total} = N A_f + A_b$$

$$25 * (0.002m\,\pi) * 0.05m + (0.03m)^2$$

$$A_{total} = 0.0087\,m^2$$

$$1 - \frac{0.0078\,m^2}{0.0087\,m^2}(1-0.95) = \boxed{0.955 = \eta_o}$$

$$R_o = \frac{1}{\eta_o h A_{total}} = \frac{1}{0.955 \times 10\frac{W}{m^2 K} * 0.0087\,m^2}$$

$$\boxed{R_o = 12.03\,\frac{K}{W}}$$

$$q_{total} = \frac{1}{R_o} * \theta_b = 8.31\ W = q_{total}$$

$$q_f = \eta_f N h A_f \theta_b = 0.95 * 25 * 10 \frac{W}{m^2 K} * 0.002 m \pi 0.05 m * 100$$

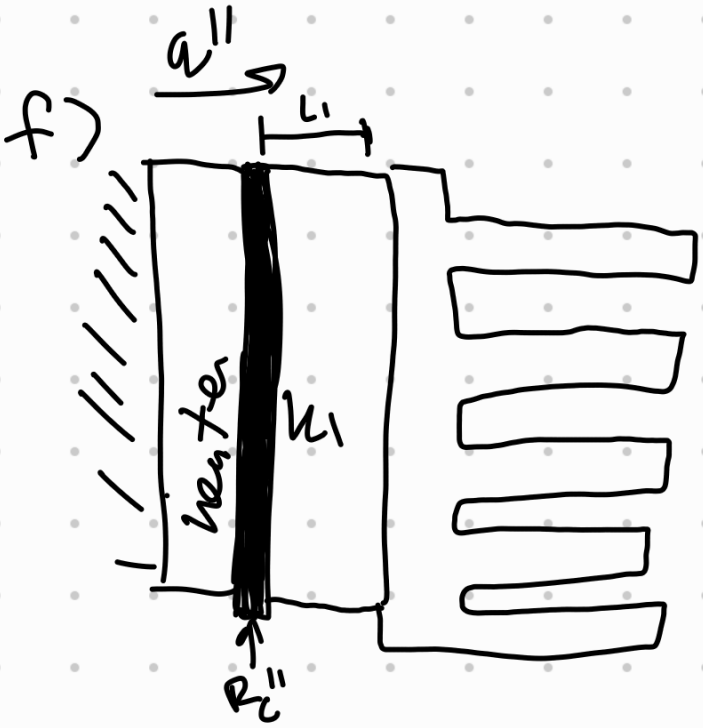
$$q_f = 7.46\ W$$

$$q_{base} = q_{total} - q_f = h A_{base} \theta_b$$

$$\boxed{q_{base} = 0.85 W}$$

$$100 \cdot 10 \frac{W}{m^2 K} * (0.03 m^2 - 25 * 0.00 m^2 \pi)$$
$$= 0.82 W$$

f)



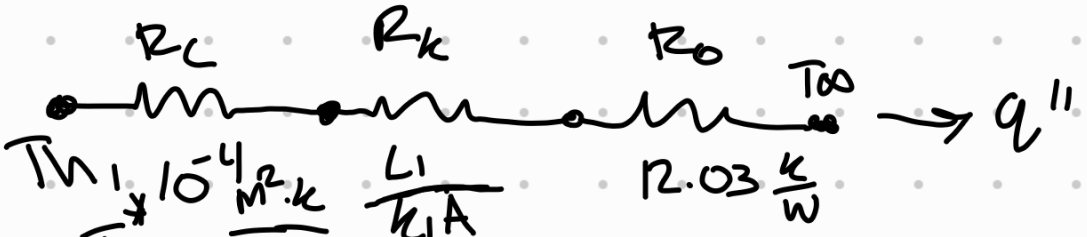
$q'' = 1000 \frac{W}{m^2}$

$L_1 = 2cm$

$R_c = 10^{-4} \frac{m^2 \cdot K}{W}$

$k_1 = 10 \frac{W}{m \cdot K}$

Draw resistance network for the system.



$$\frac{10^{-4} \frac{m^2 \cdot K}{W}}{A} \qquad \frac{L_1}{k_1 A} \qquad 12.03 \frac{K}{W}$$

$$\frac{1}{(0.03m)^2 * 10^{-4}} \rightarrow \frac{1}{9} \frac{K}{W} \qquad \frac{0.02m}{10 \frac{W}{m \cdot K} * (0.03m)^2} = 2.22 \frac{K}{m}$$

$$R_{total} = \frac{1}{9} + 2.22 + 12.03 = 14.36 \frac{K}{W}$$

$$\Delta T = q R_{total} = 1000 \frac{W}{m^2} * 14.36 \frac{K}{W} * (0.03m)^2$$

$$T_h - \overset{25°C}{T_\infty} = 12.927°C \rightarrow \boxed{T_h = 37.93\ °C}$$

Without heat sink:

$$R_{total} = \frac{1}{9} + 2.22 = 2.33 \frac{K}{W}$$

$$\Delta T = 1000 \frac{W}{m^2} * 2.33 \frac{K}{W} * (0.03m)^2$$

$$T_h - 25°C = 2.1°C$$

$$\boxed{27.1°C = T_h}$$

12.927°C w/ heat sink
vs 2.1°C w/o heat sink.